

Bharat Govil

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EDUCATION

University of Massachusetts Amherst
Amherst, MA

M.S. in Computer Science, GPA: 4.0/4.0
Expected 2027

Princeton University
Princeton, NJ

B.S.E. in Computer Science
2022

Princeton Certificates: Robotics, Linguistics, and Cognitive Science

Selected Graduate Coursework: Robotics, Generative AI, Neural Networks, Information Retrieval, AI Alignment

RESEARCH EXPERIENCE

UMass Amherst: Human-Centered Robotics Lab

May 2026 – Present
Amherst, MA

Graduate Student Researcher

- Developing a multimodal Unitree Go2 navigation policy that fuses RGB, IMU, and goal-conditioned GPS histories through transformer-based temporal fusion and rectified flow matching.
- Built PyTorch data and training pipelines over 156 training recordings (55,079 samples) and 21 validation recordings (7,813 samples), causally aligning 15 Hz RGB, 100 Hz IMU, and 5 Hz GPS into 5 Hz contexts.
- Integrating a 5 Hz ROS 2 deployment stack with distributed encoders, bounded buffers, guarded autonomy handoff, stale-data detection, command validation, and logging for real-robot trials.

Helmholtz UFZ: Environmental Risks and Sustainability Lab

May – Aug 2021
Leipzig, Germany

Machine Learning Research Intern — Advised by Prof. Christian Kuhlicke

- Trained BERT-based classifiers to identify relevant natural-hazards research papers and applied LDA to induce abstract-level research topics.
- Analyzed the temporal evolution of research topics using t-SNE embeddings and graph-based visualizations.

Princeton University: ZERO Lab

May – Sep 2020
Princeton, NJ

Data Science Research Intern — Advised by Prof. Jesse Jenkins

- Developed data-processing and optimization scripts to identify candidate electricity-generation sites from GenX system outputs.
- Created R-based geospatial visualizations of site locations and capacity distributions and documented the research workflow.

PUBLICATIONS

Leveraging Language for Accelerated Learning of Tool Manipulation

CoRL 2022

Allen Z. Ren, **Bharat Govil**, Tsung-Yen Yang, Karthik R. Narasimhan, and Anirudha Majumdar.

Proceedings of Machine Learning Research, vol. 205, pp. 1531–1541, 2023. Senior thesis advised by Prof. Karthik Narasimhan. Project page.

- Developed a language-conditioned meta-learning framework that leverages natural-language descriptions of tools (e.g., geometry and affordances) to accelerate policy adaptation in robotic manipulation tasks.
- Demonstrated significantly faster adaptation across multiple manipulation settings including pushing, lifting, sweeping, and hammering.

INDUSTRY EXPERIENCE

Genpact: Data Analytics Team

Jun 2024 – Jun 2025
New Delhi, India

Data Scientist

- Developed a monitoring service using Selenium and React with a Node.js backend to detect amendments to medical regulations across government websites.
- Integrated Azure OpenAI summarization with structured data templates, reducing client documentation effort by over 50%.
- Built a Google Custom Search API tool that reduced regulatory-document discovery time by 30% while expanding regulatory-source coverage.

Amazon Robotics: Data Lake ETL Team

Jul 2022 – Feb 2024
Boston, MA

Software Engineer

- Built AWS Kinesis applications to transform and standardize incoming robotic workcell data.
- Maintained the backend and frontend of a data lake providing centralized access to 400+ tables and streaming 400 TB of data monthly.
- Automated organization-wide access with AWS Lambda and developed health dashboards for proactive issue resolution.

RESEARCH PROJECTS

Temporally Invertible World Models via Liouville-Regularized Flow Matching Jan – May 2026

Generative AI Project

- Evaluated Liouville- and Jacobian-regularized rectified-flow world models on DMControl Walker-Walk, keeping latent rollout MSE below 1 through $H = 128$ versus over 10^{17} for unregularized JEPA-Flow.
- Implemented and evaluated the DreamerV3 baseline; co-designed experiments and analyzed rollout fidelity, cycle consistency, geometric regularization, and trajectory invertibility.

Task Separation in Language Models

Jan – May 2026

Neural Networks Project

- Probed layer-wise separation of math, riddle, and emotional-dialogue tasks across three 3B-parameter LLMs and TribeV2 using 450 balanced prompts; final-layer nearest-centroid classifiers achieved 97.3–98.0% held-out task-type accuracy.
- Led inter-model analyses of task-boundary geometry, activation-magnitude effort proxies, and temporal-to-layer alignment, finding early-layer task signal and strong Qwen–SmolLM relational agreement ($r = 0.835$).

CASE: Conflict-Aware Synthesis of Evidence

Aug – Dec 2025

Information Retrieval Project

- Designed a DSPy/Llama 3 70B pipeline that generated competing hypotheses, assigned advocate agents BM25, query-likelihood, dense, and web retrieval tools, and synthesized evidence through a legal-persona jury.
- Improved answer accuracy by 4 percentage points over prior RAG baselines and achieved a 12% gain on low-confidence decision metrics on the Bar Exam QA dataset.

Natural Language Processing: Word Sense Induction using BERT LM

Aug – Dec 2020

Junior Thesis — Advised by Prof. Christiane Fellbaum

- Integrated BERT masked-language-model substitutions with the LSDP word-sense-induction algorithm, extending the method from nouns and verbs to adjectives.
- Evaluated on SemEval-2013 Task 13, improving overall performance by 12% and adjective sense induction by 13% over prior state-of-the-art models.

TECHNICAL SKILLS

Programming: Python, C++, C, Java, Go, SQL, JavaScript

Modern Robot Learning: PyTorch, Reinforcement Learning, Imitation Learning, Flow Matching, World Models, Multimodal Learning, Meta-Learning, Vision-Language-Action Models

Classical Robotics: ROS 2 Robot Navigation, State Estimation, Localization/SLAM, Kinematics and Dynamics, PID/MPC Control

Simulation & Deployment: Isaac Sim/Isaac Lab, MuJoCo, Gazebo, CUDA, TensorRT

Language & Representation: Transformers/BERT, RAG, PCA, SVCCA, representational similarity analysis, linear probing

Data & Platforms: AWS (Kinesis, Lambda), Azure OpenAI, ETL Pipelines, Linux, Git, Docker

Languages: English, Hindi